TEXORA · LECTURE SHEET

Nonwoven Fabrics

The third great fabric family: no yarn stage at all. A web of fibres or filaments is formed and bonded directly — by needles, water jets, heat or chemistry — at speeds and costs weaving and knitting cannot approach, for products from wipes to geotextiles.

FM-4201 · ADVANCED KNITTING AND NON-WOVEN · TOPIC 3 · FABRIC MANUFACTURING SPECIALIZATION

1. Definition and classification

A **nonwoven** is a sheet of fibres or filaments assembled as a **web** and bonded into a fabric **without interlacing or interlooping yarns** — the yarn-making stage is skipped entirely. Every nonwoven process is a choice from two menus:

WEB FORMATION	PRINCIPLE	TYPICAL WEBS
Drylaid — carded	Staple fibres opened and carded into a parallel/cross-lapped web	Uniform light–medium webs; cross-lapper evens direction
Drylaid — airlaid	Fibres suspended in air, deposited on a screen	Random, bulky, absorbent webs (short fibres, pulp blends)
Wetlaid	Fibres dispersed in water, formed on a wire (paper-machine principle)	Very uniform thin webs; glass and synthetic fibre papers
Polymer-laid: spunbond	Polymer extruded as continuous filaments, drawn, laid as web in-line	Strong continuous-filament webs, medium weights
Polymer-laid: meltblown	Extruded filaments blasted by hot air into microfibres, self-bonding on the collector	Very fine fibres, low strength alone — the filtration layer

BONDING	MECHANISM	CHARACTER OBTAINED
Needle-punching (mechanical)	Barbed needles drive fibres vertically, entangling the web in Z	Bulky, strong-in-Z, no chemistry; felts, geotextiles
Hydroentangling (spunlacing)	Fine high-pressure water jets entangle fibres	Soft, drapeable, binder-free; wipes and medical

Thermal bonding	Bicomponent/low-melt fibres or powder melted at points (calender/through-air)	Clean, soft-to-firm by pattern; hygiene coverstock
Chemical bonding	Latex/resin sprayed, printed or saturated and cured	Stiff to flexible by resin; cheap and versatile
Stitch-bonding	A warp-knitted chain stitch sewn through the web	The bridge to knitting: web given yarn-level integrity

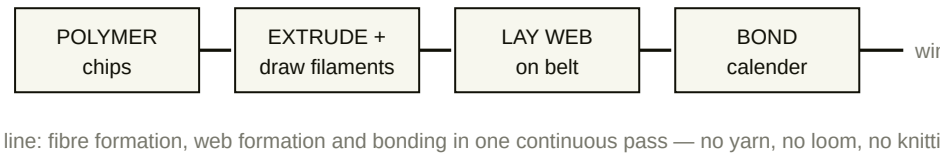


Figure 1 — Spunbond line flow (schematic). Polymer-laid lines fuse all three nonwoven stages into one machine train, which is why their economics differ so completely from woven and knitted production.

2. Production economics and a worked line calculation

Nonwovens win where the product is **area**: the line's output is web width × speed × areal mass, with no yarn packages, no warping, no sizing and no loom stops for weft. Bonding energy and raw fibre are the main costs; labour per square metre is the lowest of the three fabric families.

WORKED EXAMPLE — OUTPUT OF A NEEDLE-PUNCHED LINE

Web areal mass 300 g/m², working width 2.0 m, line speed 8 m/min.

Output = 300 g/m² × 2.0 m × 8 m/min × 60 min = 300 × 2.0 × 480 g/h ≈ **288 kg/h**.

Run the same web mass through the woven route — yarn spinning, warping, sizing, weaving, finishing — and the contrast explains the industry: nonwovens took every application where the service asked for area, bulk or filtration rather than for drape, recovery and a long service life.

3. Properties — honest comparison

PROPERTY	NONWOVEN BEHAVIOUR	REASON IN THE STRUCTURE
Tensile/tear	Moderate; anisotropic in carded webs (stronger in machine direction)	Fibres held by entanglement/bond points, not by continuous interlaced yarns; web orientation from formation
Drape/handle	Usually stiffer, paper-ward; hydroentangled types much softer	Bond points restrict fibre movement; jet entanglement leaves fibres freer

Bulk/absorbency	Excellent (airlaid, needle-punched)	High void volume, controlled pore structure
Filtration	Excellent, and designable (meltblown fineness)	Tortuous fine pores through the thickness — depth filtration (FM-4103)
Durability/wash	Many are limited-life or disposable; technical types (geotextiles) are engineered long-life	Bond points fatigue and fibres shed; durability must be specified per product
Elastic recovery	Low unless elastomeric fibres/webs designed in	No loop or crimp reservoir to recover from

4. End-uses

Hygiene: diaper and sanitary coverstock, wipes (hydroentangled). **Medical:** surgical gowns, drapes and masks (spunbond–meltblown–spunbond "SMS" composites — barrier by architecture), dressings. **Filtration:** air and liquid filter media (FM-4103 Topic 1). **Geotextiles:** needle-punched separation, drainage and protection layers in roads and landfills. **Home/industrial:** interlinings (the fusible nonwovens inside garment collars and fronts), carpet backing, wipes and polishing cloths, insulation and acoustic felts, agriculture crop cover, automotive headliners and insulation. The through-line: wherever the specification is a *function per square metre* — barrier, absorbency, filtration, separation — and not a garment's decade of flexing, the nonwoven is the default candidate.

Glossary

Web — the unbonded fibre sheet formed before bonding. **Spunbond / meltblown** — polymer-laid webs of continuous / micro-fine filaments. **Needle-punching** — mechanical Z-entanglement by barbed needles. **Hydroentangling** — entanglement by high-pressure water jets. **SMS** — spunbond–meltblown–spunbond laminate for barrier uses. **Cross-lapping** — folding a carded web back and forth to build weight and balance direction.

Key points

• Nonwoven = web formation + bonding, no yarn stage; classify any product by its pair (e.g. carded + needle-punched, spunbond + thermal).

• Line output is width × speed × gsm (worked: 288 kg/h) — area economics that weaving/knitting cannot match.

• Properties follow bonding: entanglement and bond points give bulk, filtration and absorbency; drape, recovery and long fatigue life are the structural weaknesses.

• SMS-type composites show the family's design style: layers specialised by function, like the laminates of FM-4103.

Practice

1. Specify formation + bonding for (a) a baby wipe, (b) a road geotextile, (c) a respirator filter layer, justifying each in one line.
2. A spunbond line of width 3.2 m runs 250 m/min at 25 g/m². Compute its output in kg/h.
3. Why are carded webs anisotropic, and which formation step reduces the anisotropy?
4. Give two garment functions for which nonwovens have NOT displaced wovens/knits, with the structural reason.

Answer key

1. (a) **Airlaid/carded + hydroentangled:** soft, absorbent, binder-free against skin. (b) **Carded/cross-lapped + needle-punched:** thick, Z-entangled strength and drainage for soil contact. (c) **Meltblown (polymer-laid, self-bonded):** microfine fibres give the fine tortuous pores that capture particles.
2. Output = 25 × 3.2 × 250 × 60 ÷ 1000 = **1200 kg/h**.
3. Carding lays fibres preferentially along the machine direction, so strength and extension differ MD vs CD. **Cross-lapping** folds the web back and forth at an angle, stacking layers whose orientations partly cancel and building a more balanced, heavier web.
4. (i) **Tailored outerwear shells:** need drape, crease recovery and years of flexing — bond-point structures fatigue and handle boardy. (ii) **Stretch-fit garments:** need the loop structure's extension and recovery reservoir; a bonded web has no comparable low-force recovery mechanism.

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